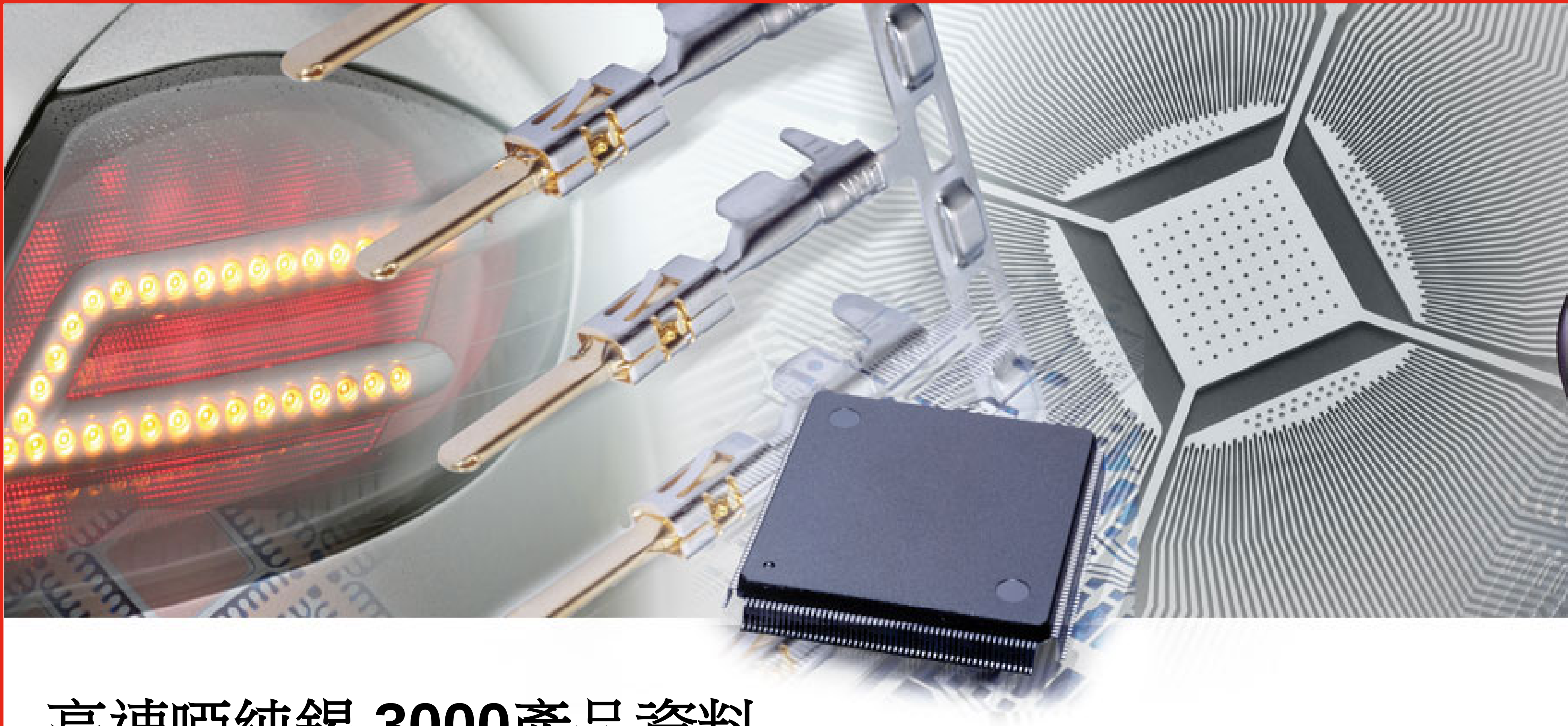


高速哑純錫 3000產品資料
StannoPure 3000
Product Information

StannoPure 3000
產品資料

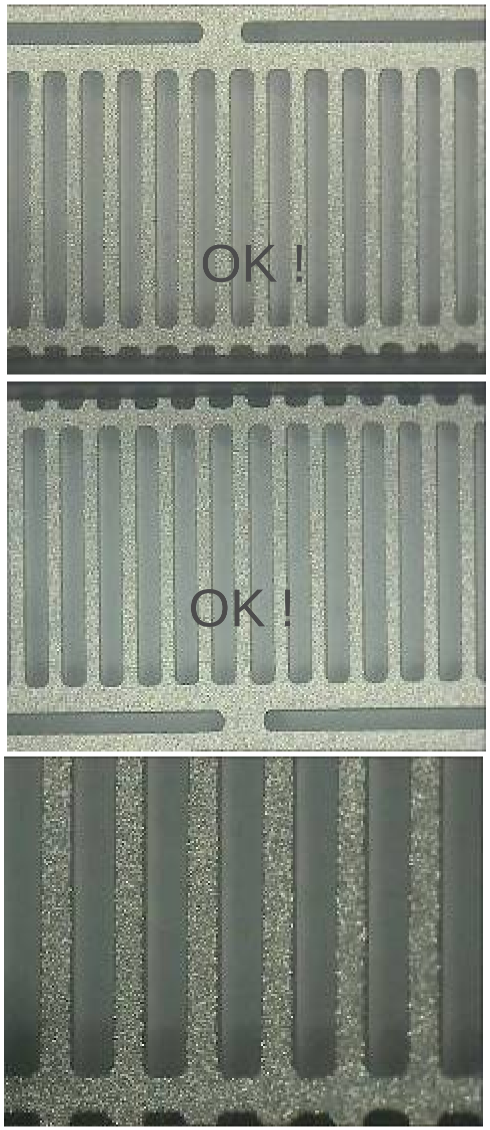
StannoPure 3000

StannoPure 3000

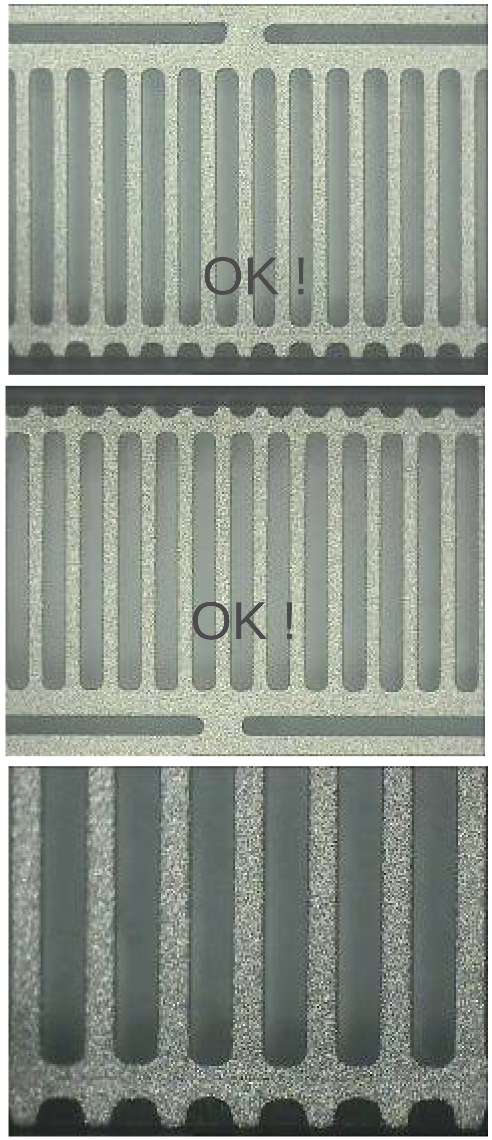
電鍍產品: LQFP 100 – 2 塊陽極

錫含量 60 g/L ; 溫度 45℃ ; 20A ; 時間 1 分鐘

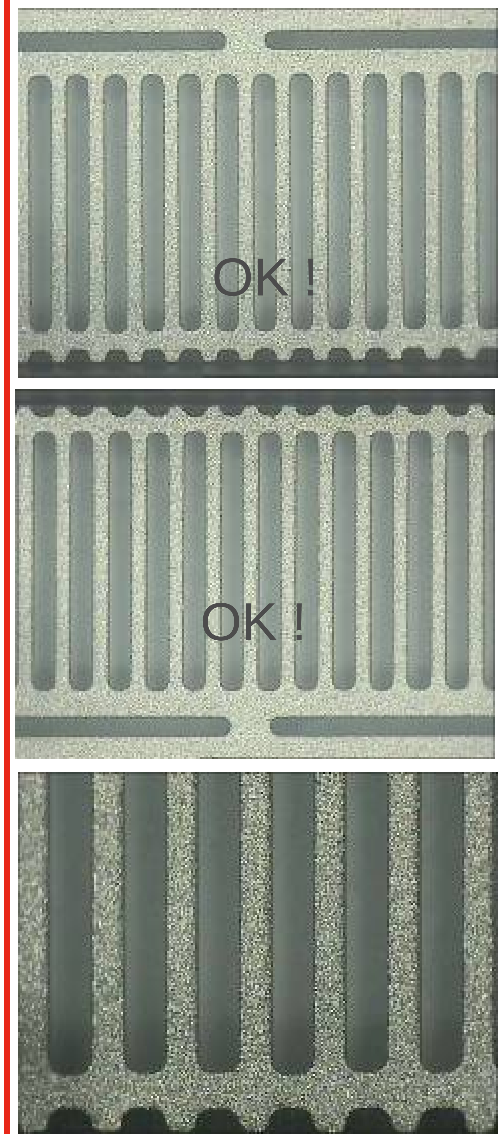
游離酸 100



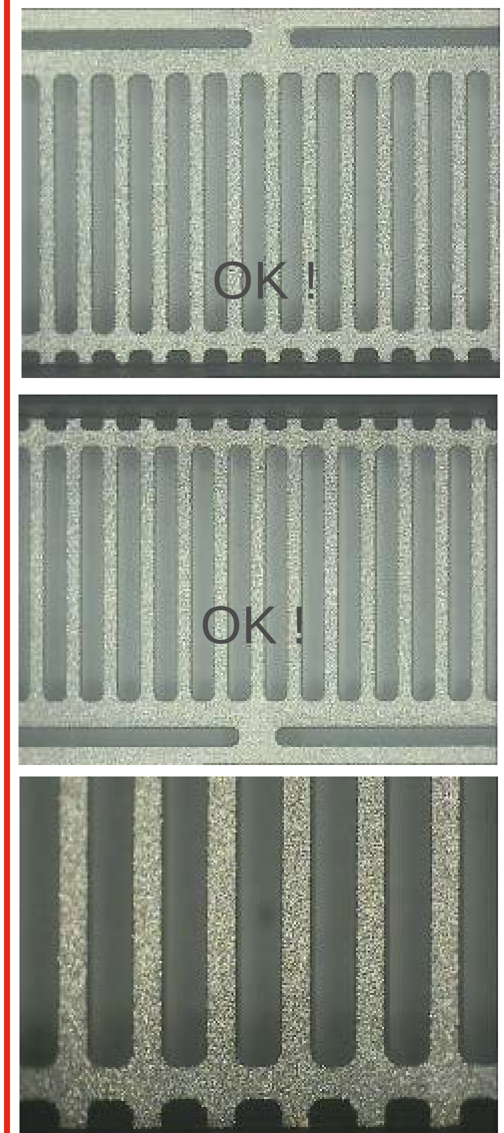
游離酸 150



游離酸 200



游離酸 250



StannoPure 3000

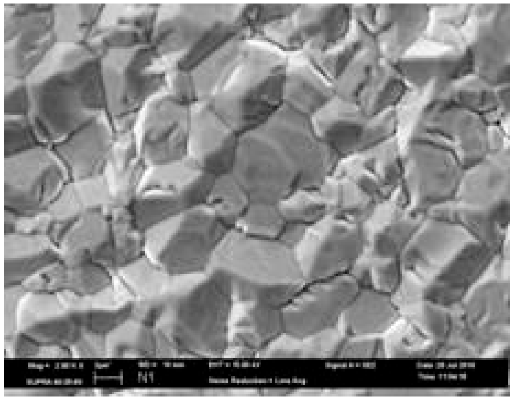
ATOTECH

StannoPure 3000

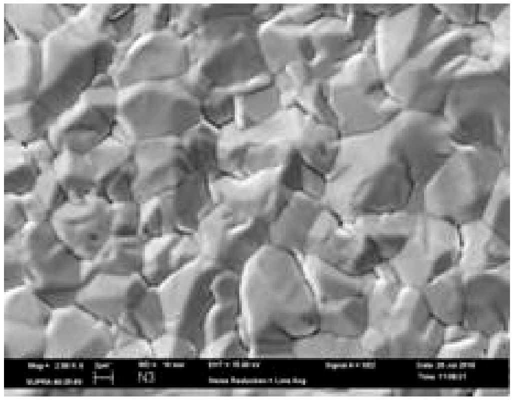
電鍍產品: LQFP 100 – 2 塊陽極

10 微米 – 20 ASD

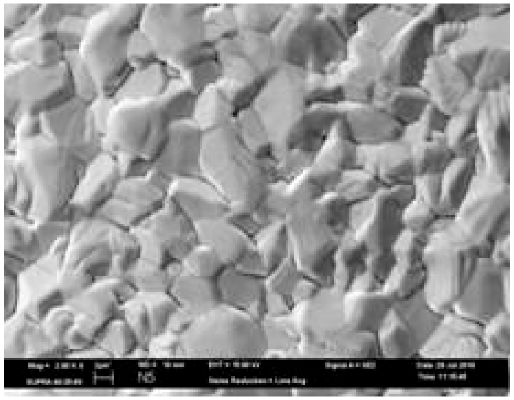
錫 60g/L; 酸 100



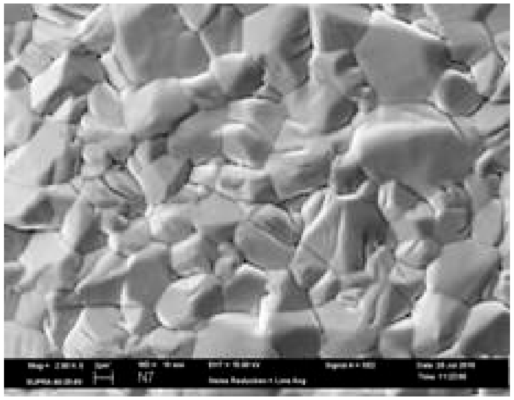
錫60g/L;酸150



錫60g/L;酸200

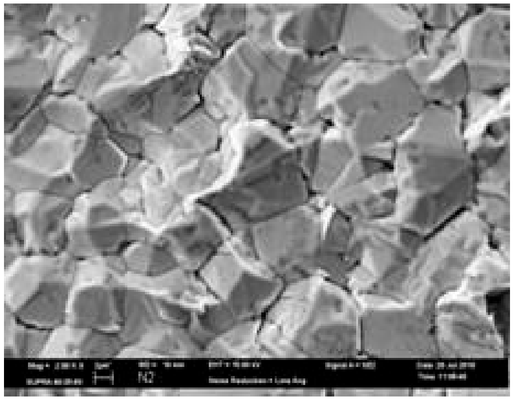


錫60g/L;酸250

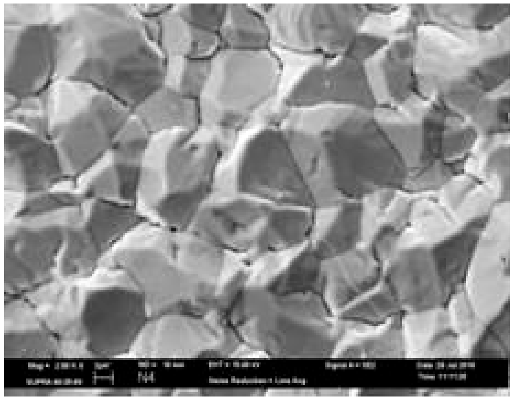


15 微米 – 30 ASD

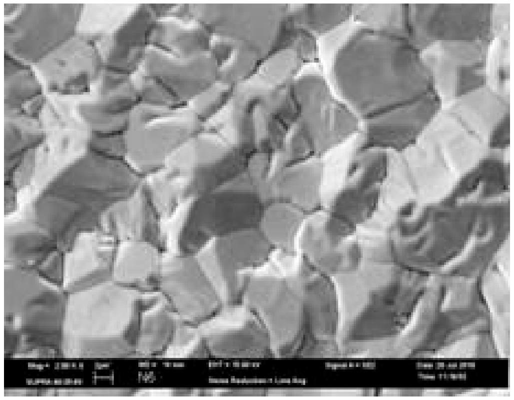
錫60g/L;酸100



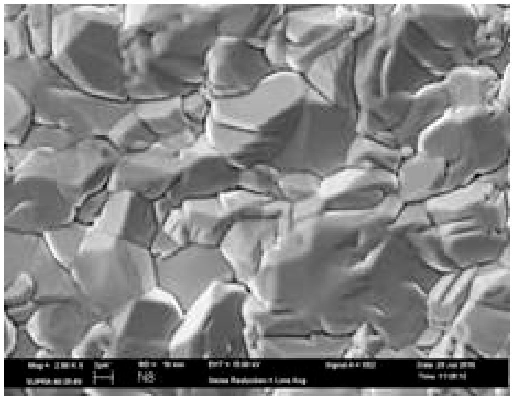
錫60g/L;酸150



錫60g/L;酸200



錫60g/L;酸250



StannoPure 3000

侯氏槽測試

增加錫濃度

55g/L 錫 | 65g/L錫 | 75g/L錫 | 85g/L錫

增加酸濃度

100g/L酸

150g/L酸

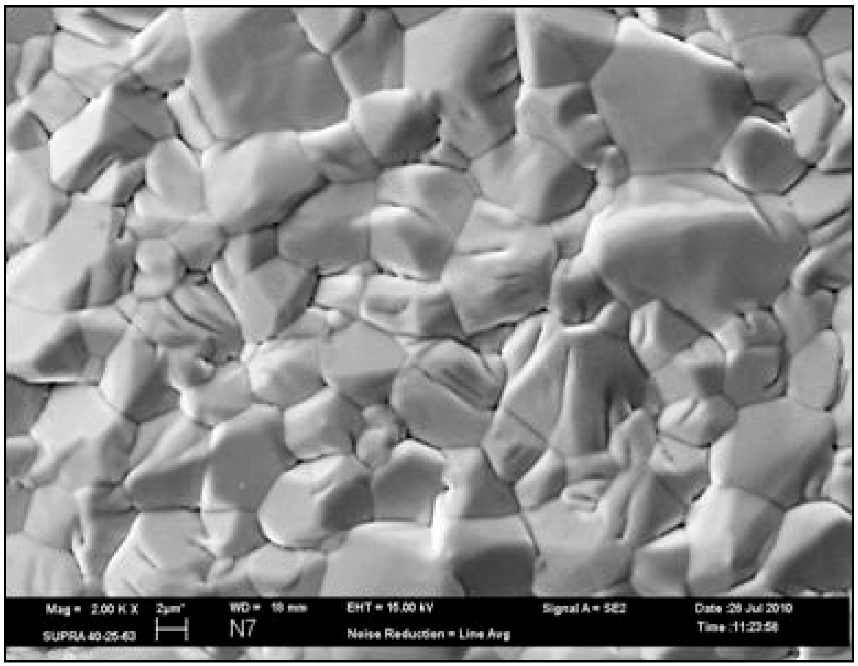
200g/L酸

250g/L酸

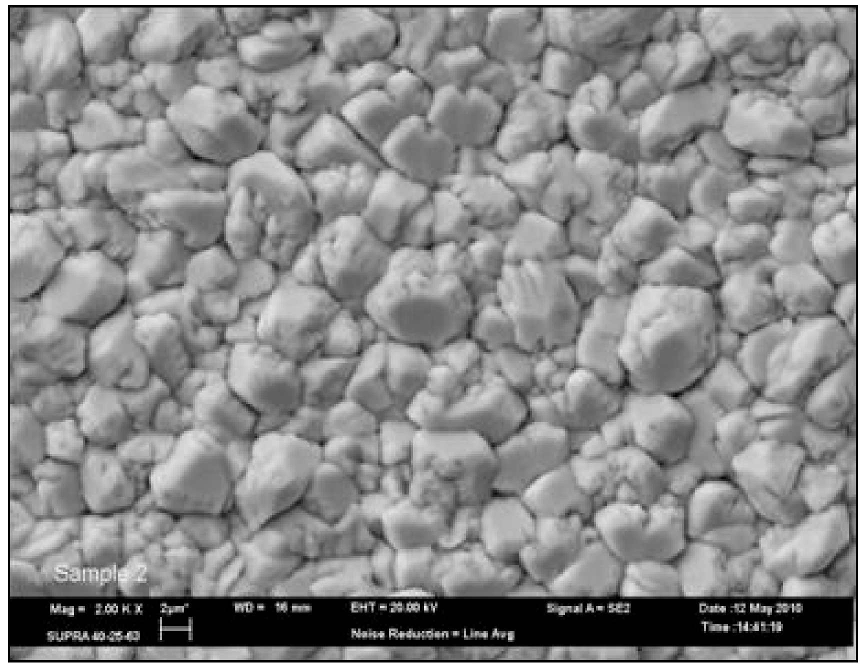
鍍層形態 (1)

StannoPure 3000 Vs StannoPure HSM HT/ST(在高游離酸下)

錫 60 g/L, 酸 250 g/L,
10 微米, 20ASD



StannoPure 3000

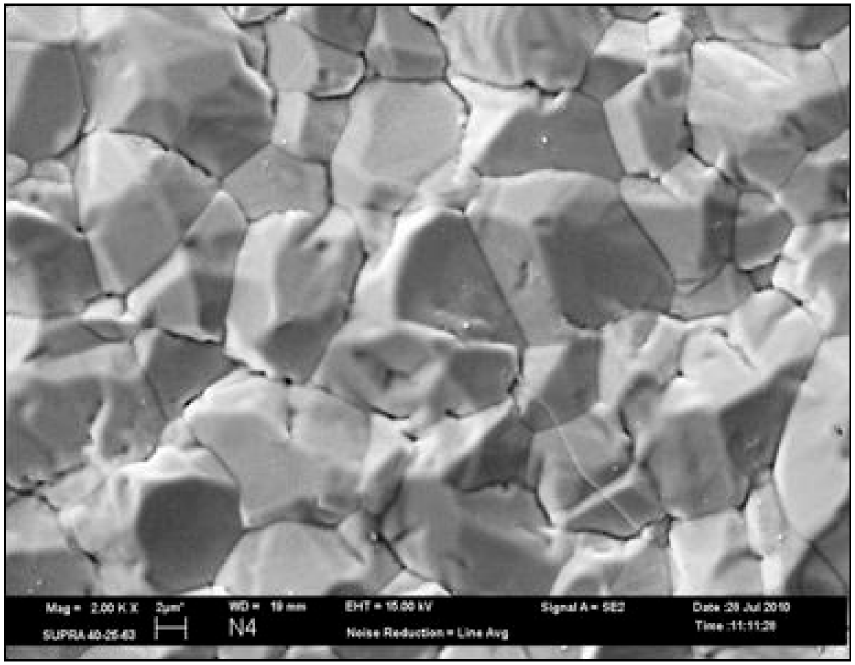


StannoPure HSM HT/ST

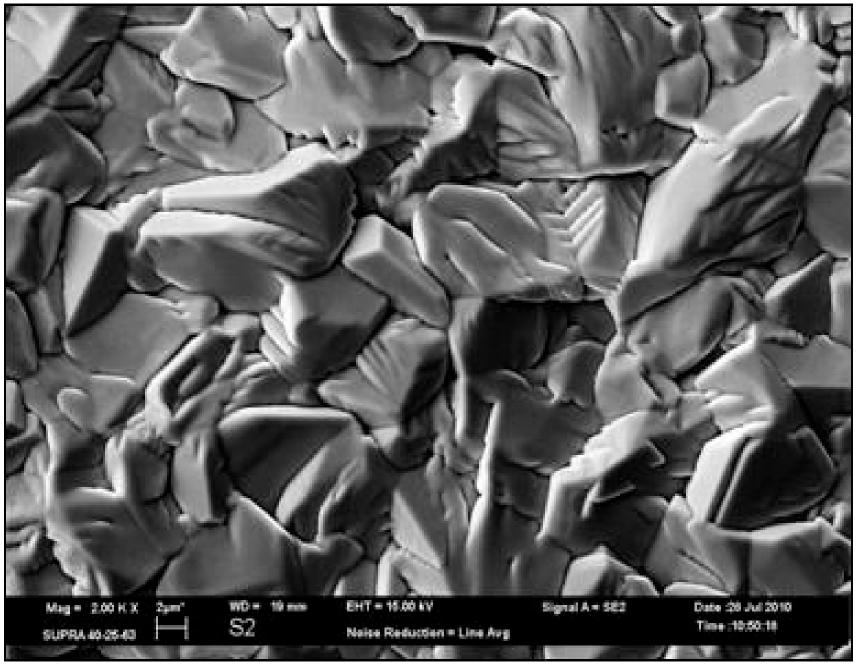
鍍層形態 (2)

StannoPure 3000 Vs 競爭對手 S 在高電流密度

錫 60 g/L, 酸 250 g/L,
15 微米, 30ASD



StannoPure 3000

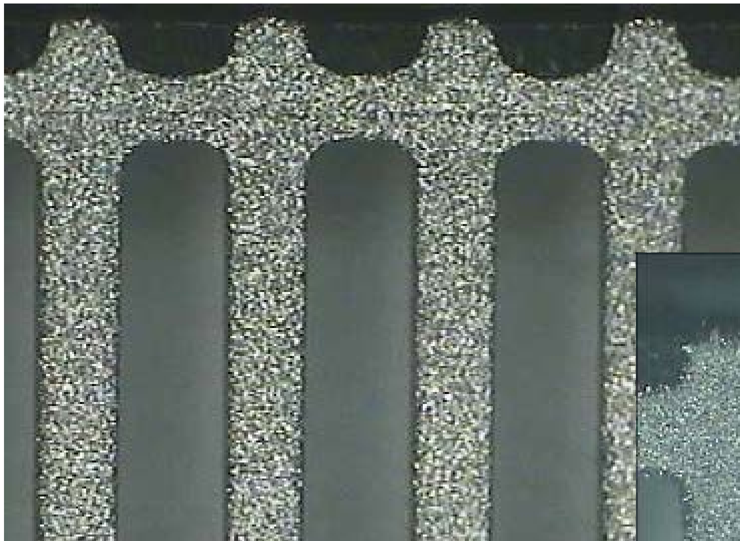


競爭對手 S

StannoPure 3000 比較平滑

樹枝狀晶體比較

StannoPure 3000



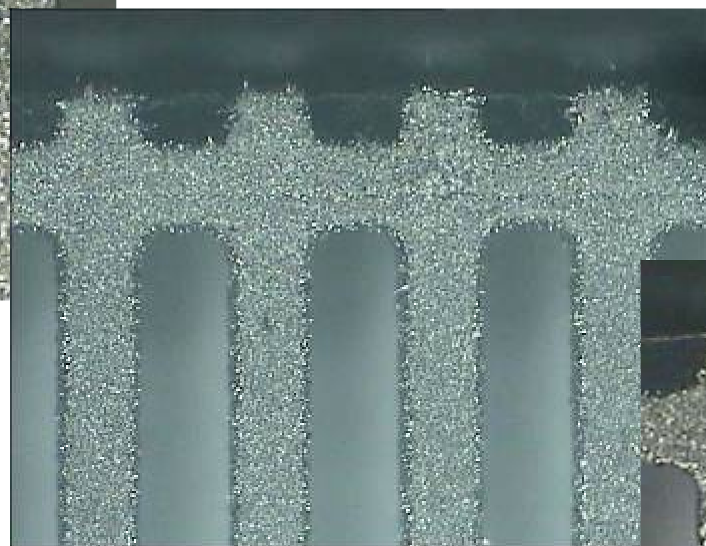
錫 60 g/L

酸 150g/L

LQFP

電流= 20A

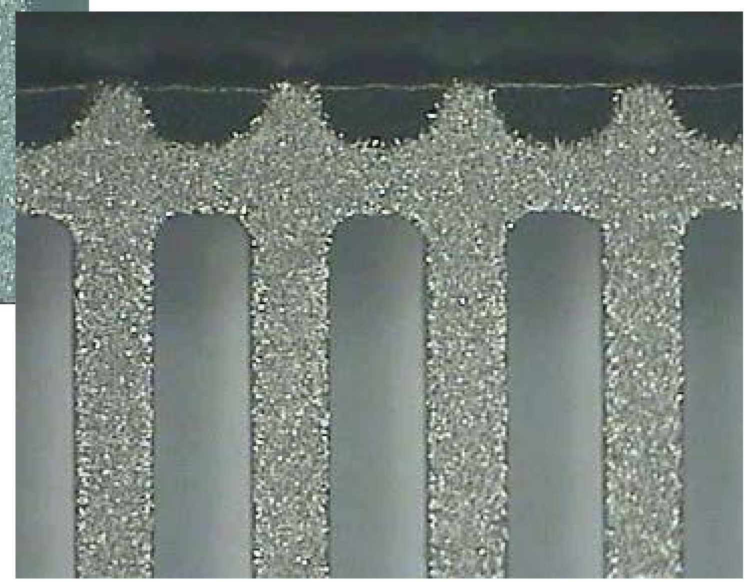
Vs HSM HT/ST



錫 70 g/L

酸 150g/L

Vs 競爭對手 S



錫 60 g/L

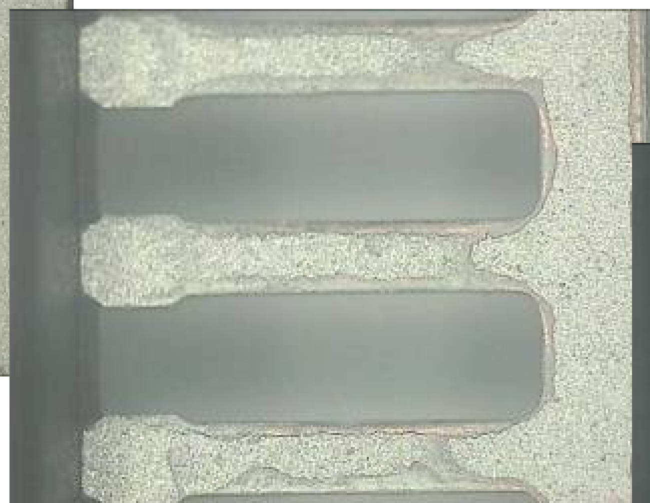
酸 150g/L

StannoPure 3000 的鍍層上沒有發現樹枝狀晶體

不均勻鍍層 / 漏鍍比較

StannoPure 3000

改良版NiveoStan



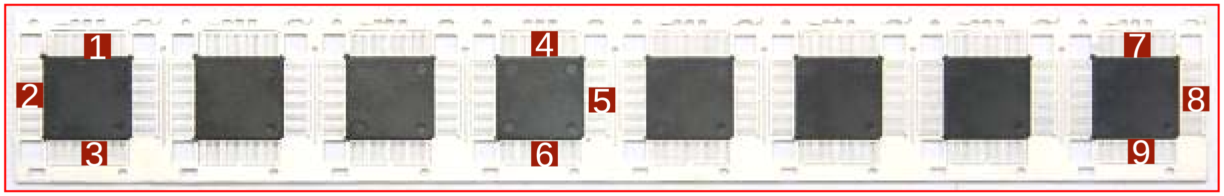
競爭對手 S

StannoPure 3000 及競爭對手 S 皆沒有不均勻鍍層

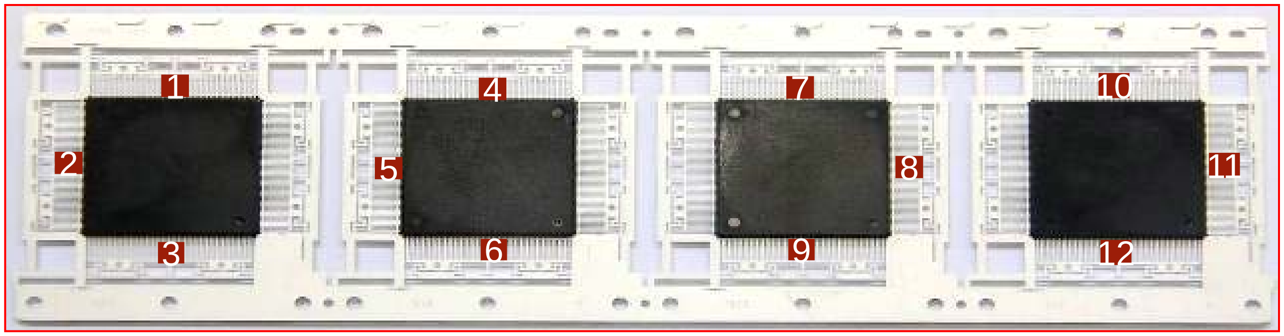
StannoPure 3000

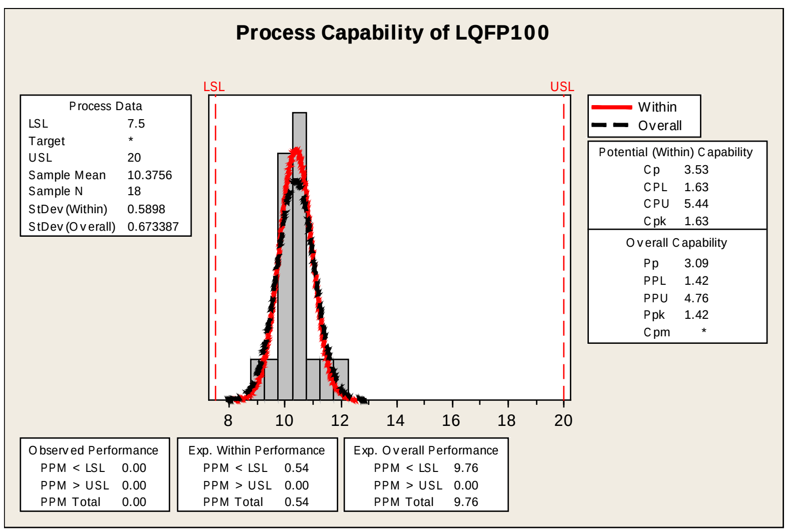
厚度分佈

LQFP 100

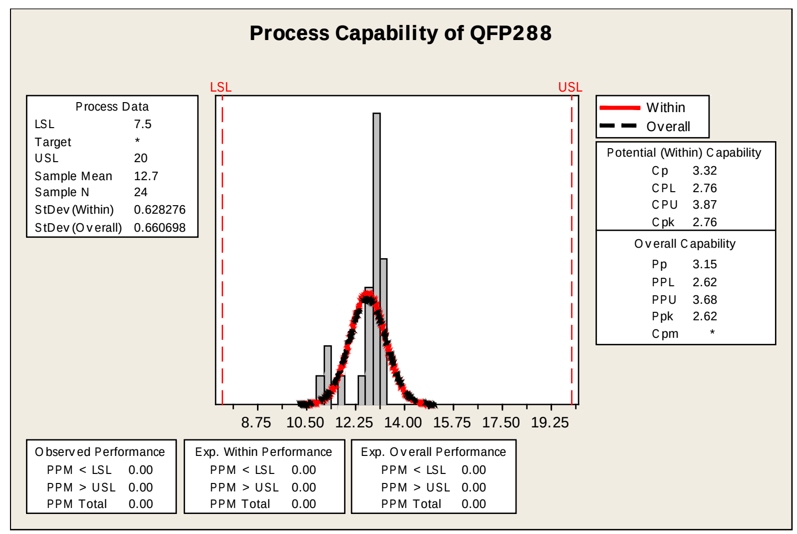


LQFP 288





Cpk = 1.63

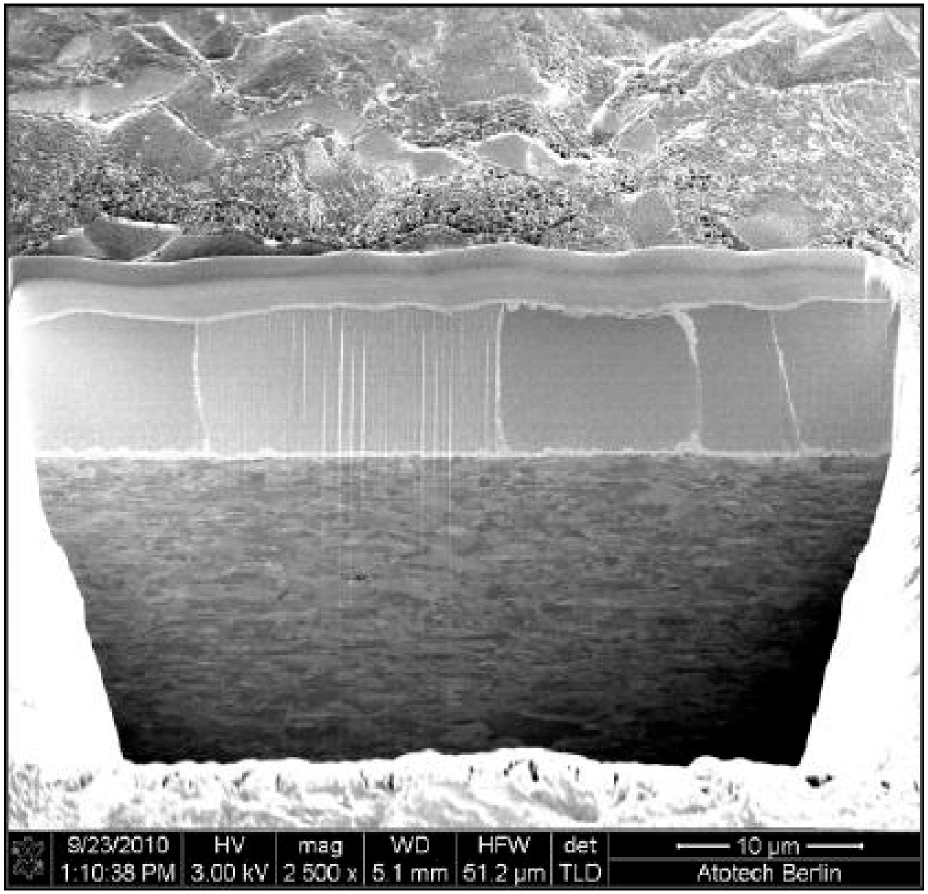


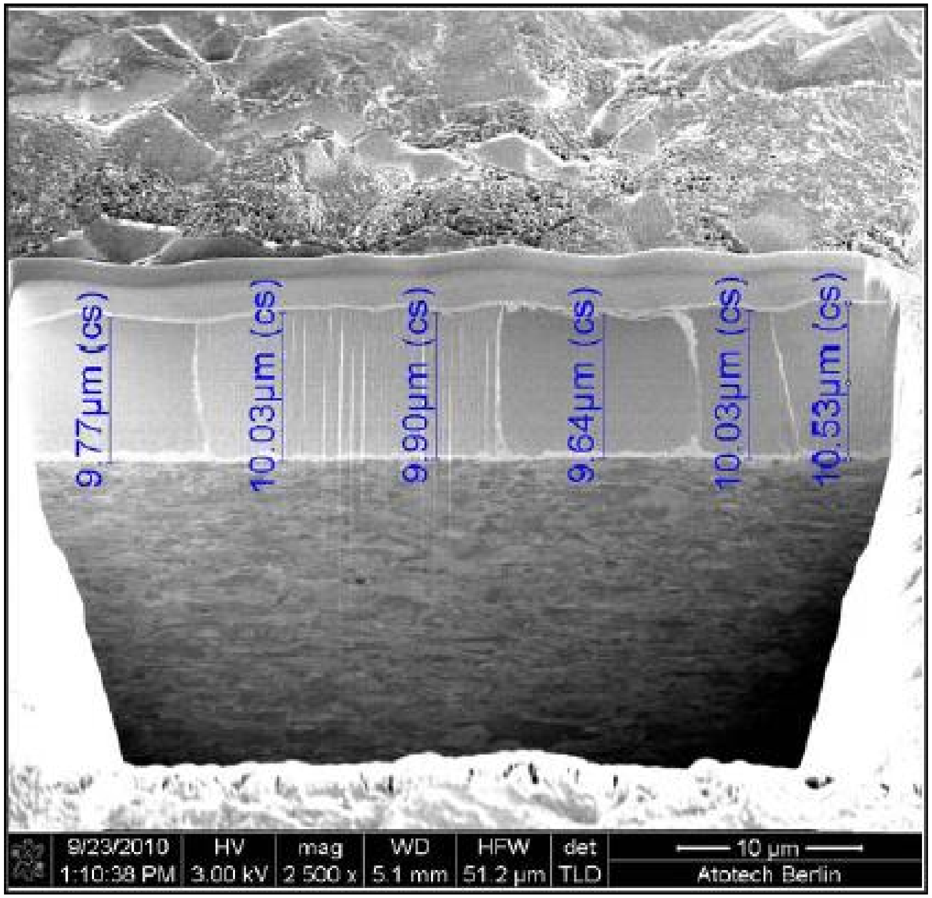
Cpk = 2.76

- 理想的Cpk值

StannoPure 3000

大晶體 - HSOP 20L(20ASD)

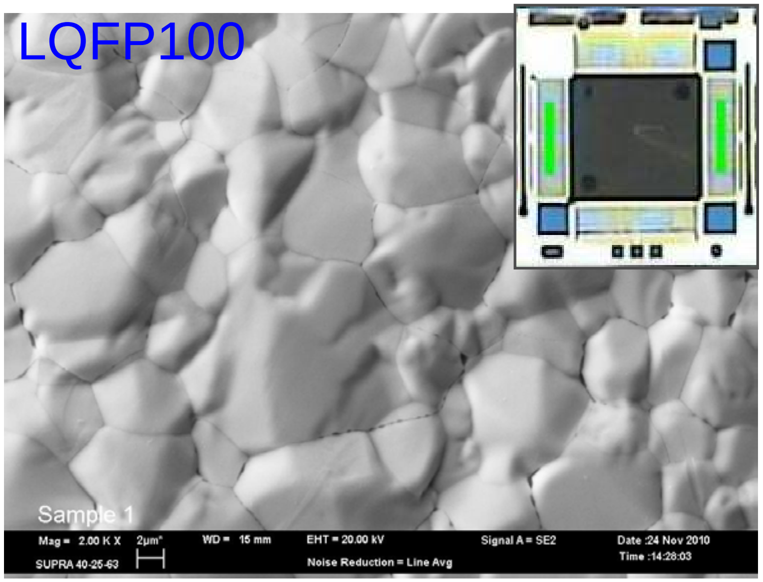


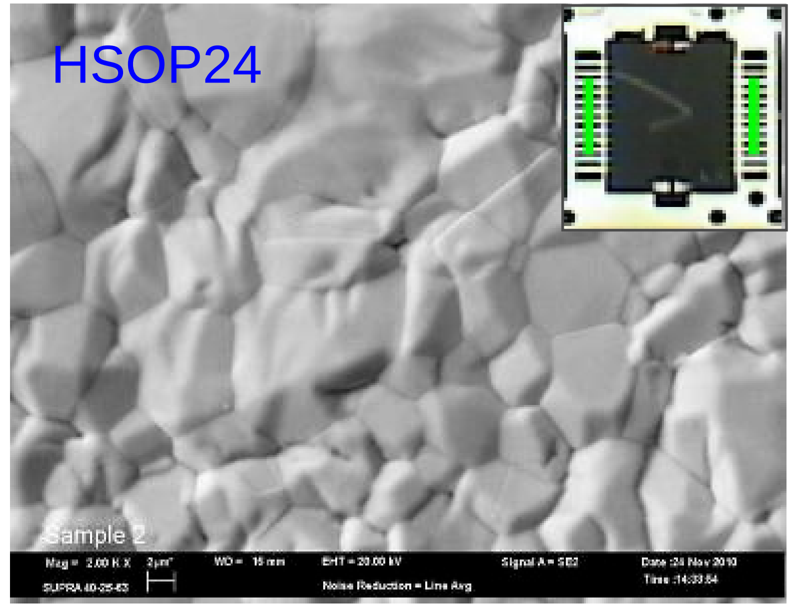


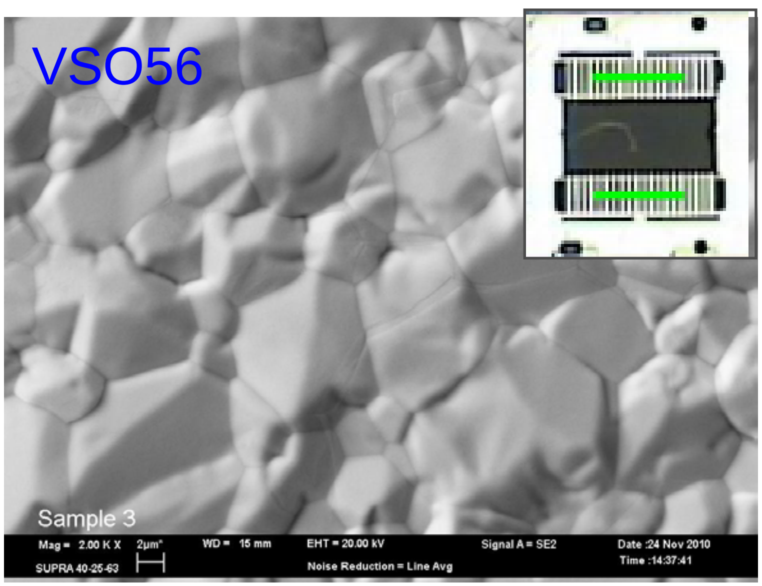
- 大晶體及非常平滑的鍍層

StannoPure 3000

SEM – 平滑形態







- 鍍層非常平滑

StannoPure 3000

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RSAI

1
2
3

S_A (nm)							
Sample	Measuring window	Raw data					Average values
1	(124,8 µm)²	473	437	452	466	446	455
	(62,4 µm)²	423	458	432	438	444	439
2	(124,8 µm)²	468	483	481	453	486	474
	(62,4 µm)²	472	448	462	463	433	455
3	(124,8 µm)²	401	406	409	423	463	420
	(62,4 µm)²	311	320	305	305	297	307

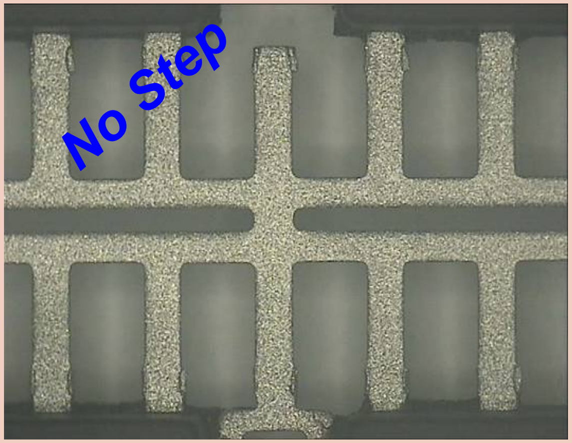
RSAI (%)							
Sample	Measuring window	Raw data					Average values
1	(124,8 µm)²	14,3	14,4	14,5	15,4	14,0	14,5
	(62,4 µm)²	17,9	20,6	20,5	19,8	21,0	20,0
2	(124,8 µm)²	15,8	16,0	14,6	16,4	15,1	15,6
	(62,4 µm)²	23,6	21,8	18,7	21,9	21,9	21,6
3	(124,8 µm)²	13,6	12,2	12,1	14,0	12,6	12,9
	(62,4 µm)²	16,2	17,2	15,1	17,3	16,6	16,5

- 能達到低RSAI值

StannoPure 3000

ATOTECH

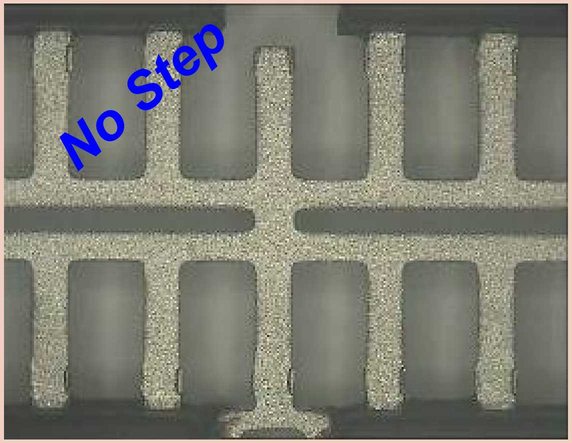
不均勻鍍層



20ASD

15ASD

10ASD



5ASD

StannoPure 3000

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穩定性測試

開始

沒有懸浮分解物
(100˚C)

沒有懸浮分解物
(2小時後@
100˚C)

- 當鍍液加熱至**105ºC**時，仍沒有變混濁(濁點)
- 經過**2**小時**100ºC**加熱後，沒有發現懸浮分解物

StannoPure 3000

ATOTECH

單一添加劑

成份	新開缸	10hrs電解	20hrs電解
Comp 1	101%	87%	81%
Comp 2	102%	90%	80%
Comp 3	100%	87%	84%

- 主要成份比例沒有因**20**小時電解而失控
- 無需擔心成份比例失調
- 只需分析**1**種成份

註： *假設,*

生產線電流 = 70A x3 + 60A x 2 = 330A

體積 – 950 Lt

每天生產時間 = 18 hrs

Ahr / 升 / 日 = 5.21

所以, 200Ahr / 升相等於38.4日

StannoPure 3000

侯氏片外觀

新開缸

10小時電解

20小時電解

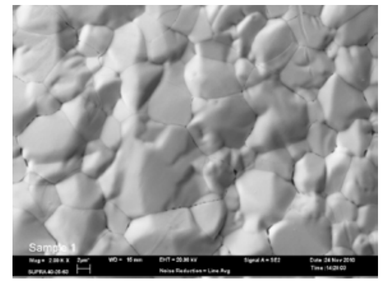
- 經過20小時電解亦無

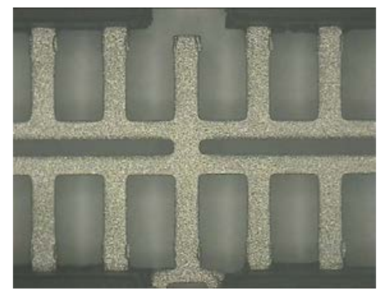
StannoPure 3000

ATOTECH

適合用於高速鋼帶機上

- 良好的分佈
- 大晶體，平滑形態鍍層，低**RSAI**
- 單一添加劑，容易控制
- 沒有不均勻鍍層
- 沒有懸浮分解物





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Thank You !